

Python Project: Face Detection System for a Digital Camera

Project Description:

I developed a Face Detection system for a consumer compact digital camera (ProCam S.p.A.) aimed at improving the automatic camera settings for single- and multi-person selfies. The system receives a photo and returns a list of bounding-box coordinates for all detected faces (or an empty list if no faces are present). The challenge was to build a robust, low-compute face detector from scratch using classical machine-learning techniques (Scikit-learn), without relying on pre-trained deep models, and to make it suitable for deployment on resource-constrained hardware.

Key Responsibilities and Tasks

I led the end-to-end pipeline: sourcing and preparing training data (public face datasets and custom-collected images, plus manual annotation where needed), defining preprocessing steps (grayscale conversion, normalization, multi-scale image pyramid), and extracting lightweight image descriptors (e.g., HOG / gradient-based features). I implemented a sliding-window detector and trained linear classifiers with Scikit-learn (SVM / logistic regression), including iterative hard-negative mining and cross-validation to improve generalization. I measured performance with standard detection metrics (precision/recall, IoU-based matching) and optimized the model for low memory/CPU usage through model simplification, feature dimensionality reduction, and quantized serialization. Finally, I integrated the detector into a minimal runtime (OpenCV for image handling, a small wrapper that returns bounding-box lists) and produced full documentation explaining algorithmic choices, preprocessing, training protocol, and evaluation.

Outcome

The resulting pipeline reliably returned face bounding boxes (or an empty list when appropriate) and identified typical selfie scenarios (single face, multiple close faces) with competitive precision while remaining lightweight enough for near-real-time use on limited hardware. The approach improved the camera's auto-configuration for portrait/selfie shots, reducing manual adjustments by users and enabling more consistent exposure/framing in multi-person shots. All experiments, failure cases, and performance trade-offs were documented to inform the next product iteration.

Technologies and Tools Used

Scikit-learn (model training and evaluation), scikit-image / OpenCV (image I/O, HOG feature extraction, preprocessing), NumPy, Pandas (data handling), joblib (model persistence), standard annotation tools (for custom dataset labeling), and common evaluation tooling (IoU, precision/recall scripts). In the project report I also reference classical face-detection literature (e.g., HOG+SVM approaches) and recent practical guides used to design and justify the pipeline.